

F8NET: FIXED-POINT 8-BIT ONLY MULTIPLICATION FOR NETWORK QUANTIZATION

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ABSTRACT

Neural network quantization is a promising compression technique to reduce memory footprint and save energy consumption, potentially leading to real-time inference. However, there is a performance gap between quantized and full-precision models. To reduce it, existing quantization approaches require high-precision INT32 or full-precision multiplication during inference for scaling or dequantization. This introduces a noticeable cost in terms of memory, speed, and required energy. To tackle these issues, we present F8Net, a novel quantization framework consisting of only fixed-point 8-bit multiplication. To derive our method, we first discuss the advantages of fixed-point multiplication with different formats of fixed-point numbers and study the statistical behavior of the associated fixed-point numbers. Second, based on the statistical and algorithmic analysis, we apply different fixed-point formats for weights and activations of different layers. We introduce a novel algorithm to automatically determine the right format for each layer during training. Third, we analyze a previous quantization algorithm—parameterized clipping activation (PACT)—and reformulate it using fixed-point arithmetic. Finally, we unify the recently proposed method for quantization fine-tuning and our fixed-point approach to show the potential of our method. We verify F8Net on ImageNet for MobileNet V1/V2 and ResNet18/50. Our approach achieves comparable and better performance, when compared not only to existing quantization techniques with INT32 multiplication or floating-point arithmetic, but also to the full-precision counterparts, achieving state-of-the-art performance.

1 INTRODUCTION

Real-time inference on resource-constrained and efficiency-demanding platforms has long been desired and extensively studied in the last decades, resulting in significant improvement on the trade-off between efficiency and accuracy (Han et al., 2015; Liu et al., 2018; Mei et al., 2019; Tanaka et al., 2020; Ma et al., 2020; Mishra et al., 2020; Liang et al., 2021; Jin et al., 2021; Liu et al., 2021). As a model compression technique, quantization is promising compared to other methods, such as network pruning (Tanaka et al., 2020; Li et al., 2021; Ma et al., 2020; 2021a; Yuan et al., 2021) and slimming (Liu et al., 2017; 2018), as it achieves a large compression ratio (Krishnamoorthi, 2018; Nagel et al., 2021) and is computationally beneficial for integer-only hardware. The latter one is especially important because many hardwares (e.g., most brands of DSPs (Ho, 2015; QCOM, 2019)) only support integer or fixed-point arithmetic for accelerated implementation and cannot deploy models with floating-point operations. However, the drop in performance, such as classification accuracy, caused by quantization errors, restricts wide applications of such methods (Zhu et al., 2016).

To address this challenge, many approaches have been proposed, which can be categorized into *simulated* quantization, *integer-only* quantization, and *fixed-point* quantization (Gholami et al., 2021). Fig. 1 shows a comparison between these implementations. For simulated quantization, previous works propose to use trainable clipping-levels (Choi et al., 2018), together with scaling techniques on activations (Jin et al., 2020b) and/or gradients (Esser et al., 2019), to facilitate training for the quantized models. However, some operations in these works, such as batch normalization (BN),

*Work done during an internship at Snap Inc. Code is available at <https://github.com/snap-research/F8Net>.